TESTING PARAMETERS WITH TEMPERATURE

Test	Material	Temperature	Testing Time	Humidity	Load	Remarks
Colorfastness to perspiration	**Lining:** Fabric, Leather, Synthetic, Sockliner, Webbing, Thread, Ink & Print, Woben Label, Elastic cord	38±2°C	6 hrs.	N/A	4.5kg	Ink & Print, Elastic Cord, Woven Label, and Webbing should be tested only if in direct contact with skin.
	Top side: Leather-Grain side only, Synthetic, Fabric, Thread, Elastic cord, Woven label, Sockliner					Applies to sockliner with or without top cloth.
	Back side: Synthetic, Woven label					Load is to be removed before placing the sample in the oven.
Colorfastness to water	**Upper:** Fabric, Leather, Synthetic, Velcro, Thread, Lace, Ink & Printed, Zipper, Woven label, Elastic cord, Webbing	38±1°C	6 hrs.	N/A	4.54kg	Ink & Print - For ink printed on upper material
	Top side: Leather, Synthetic, Fabric, Velcro, Thread, Lace, Ink & printed, Zipper,Woven label, Elactic cord, Webbing					
	Back side: Leather, Synthetic, Woven label					
Shrinkage	Rubber Outsole, Foxing Tape, Zipper	70±1 °C	1 hr. ±1 min	N/A	N/A	
	Sockliner EVA, Latex, PU, Insole	60±1 °C	1 hr. ±1 min			
Shrinkage - After Hydrolysis	Rubber Outsole	60±1 °C	7 days	95% RH	N/A	
Shrinkage - After Heat Aging	Rubber Outsole	70±1 °C	72 hrs	N/A		
Heat Aging	Fabric Upper/ Ink & Print	70°C	72 hrs	N/A		Heat Aging - white & light color and coated & printed material
	Fabric Lining, Leather Upper & Linning, Synthetic Upper & Linning, Eyelet, Webbing, Lace, Thread, Velcro, Zipper, Elastic gore, Woven label, Rubber Outsole. Sockliner - EVA, PU	70°C	72 hrs			Foxing tape - For light colors
	Foxing tape					No change on physical appearance
Hydrolysis	Fabric Upper/ Ink & Print	60°C	7 days	95% RH	N/A	Hydrolysis - white & light color and coated & printed material
						No delamination of coating or ink
	Leather Upper & Lining,					Some material analysis include color change. Review the spec.
						No delamination of coating/ Polyurethane coated leather including split leather PU coated
	Synthetic Upper & Lining					No delamination of coating/ Polyurethane coated synthetic leather including synthetic split leather PU coated
	Eyelet					80% retained of the original strength
						No peeling off coating and no deformation of material
	Rubber Outsole					80% retained of the original strength
	Foxing tape					No change on physical appearance
	Eva Midsole					80% retention of tensile strength value before hydrolysis aging / Physical appearance didn't hydrolyze
	Sockliner EVA, Latex, PU					No delamination of ink
	PU Foam					80% retention of tensile strength value before hydrolysis aging
						Physical appearance of PU didn't hydrolyze
	Webbing					No deterioration of physical appearance

Test	Material	Temperature	Testing Time	Humidity	Load	Remarks
Hydrolysis	Elastic gore	60°C	7 days	95% RH	N/A	No deterioration of physical appearance
	Woven label					
	Rubber Outsole, Foxing tape,					No change on physical appearance
Blooming test - After Hydrolysis	Rubber Outsole, Foxing tape,	60°C	7 days	95% RH	N/A	Rubber outsole and components that likely will bloom
PVC Migration	Leather Upper	50±2°C	16 hrs	N/A	1.0kg	For colored
Ultraviolet Light Discoloration	Fabric Upper & Lining	50°C	24 hrs	N/A	N/A	For light color
			12 hrs			For neon color
	Leather Upper & Lining,		24 hrs			For light color
			12 hrs			For neon color
	Synthetic Upper & Lining		24 hrs			For light color
			12 hrs			For neon color
	Webbing, Lace, Thread, Velcro, Elastic gore, Woven label,		24 hrs			For light color
			12 hrs			For neon color
	Foxing tape		24 hrs			
	Sockliner EVA, PU, PU Foam					For sockliner with or without top cloth / For PU foam - UV light test is for light colors, for exposure part. Any material which has been approved at levels below 3.5 should not be used in exposure parts.
	Zipper					For plastic element only, also for light colors tape
	Eyelet					Compound for cold weather applications should be ≥ 100 kgf/cm2
	Rubber outsole		24 hrs			For light color
			12 hrs			For neon color
Compression	Sockliner EVA, Latex, PU	45±2°C	6 hrs	N/A	N/A	SAMPLE THICKNESS: 8.0-8.5mm — SPACER THICKNESS: 4.0mm
						8.5-9.5mm — 4.5mm
	PU foam	23±2°C	22 hrs			9.5-10.5mm — 5.0mm
						10.5-11.5mm — 5.5mm
						11.5-12.0mm — 6.0mm

1. Within 15 minutes after compressing the samples should put in the oven.
2. Recovery period after test:
   - 30±2 minutes for EVA, Latex, PE & PU
   - 60±2 minutes for soft PU
3. After to recovery, should measure the final thickness.